**Bansilal Ramnath Agarwal Charitable Trust's**

# Vishwakarma Institute of Technology, Pune-37

*(An autonomous Institute of Savitribai Phule Pune University)*

## Department of Multidisciplinary Engineering

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|---|---|
| **Division** | AIML-A |
| **Batch** | 01 |
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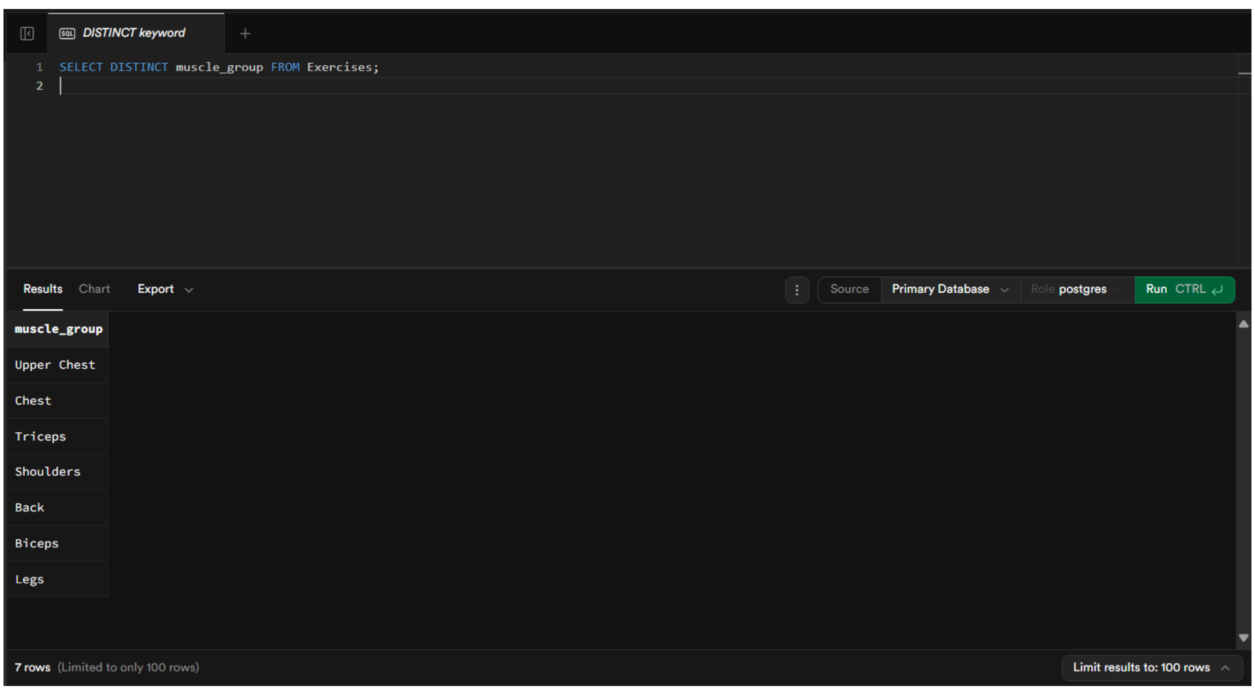
**Title:**
Distinct keyword, where clause, on clause, having clause, group by clause with aggregate functions, set operations, nested query and correlated query

**Description:** In this assignment, we explored key SQL concepts used for querying databases. The DISTINCT keyword helps remove duplicate values, while the WHERE clause filters rows based on conditions. The ON clause specifies join conditions between tables. The GROUP BY clause groups records for analysis, and with aggregate functions like COUNT or AVG, it summarizes data, while the HAVING clause filters these groups. Set operations such as UNION, INTERSECT, and EXCEPT combine or compare results of multiple queries. Nested queries (subqueries) are used inside another query for filtering or calculation, while correlated queries depend on values from the outer query for execution. Together, these concepts allow efficient data retrieval and analysis.

**Screenshots/Output:**



WHERE clause

```sql
SELECT name, muscle_group FROM Exercises WHERE muscle_group = 'Chest';
```

Results Chart Export

| name | muscle_group |
| --- | --- |
| Chest Fly | Chest |
| Push Up | Chest |
| Push Up | Chest |

3 rows (Limited to only 100 rows)

ON clause

```sql
SELECT e.name, eq.equipment_name
FROM Exercises e
JOIN Available_Equipments eq
ON e.equipment_id = eq.equipment_id;
```

Results Chart Export

| name | equipment_name |
| --- | --- |
| Bicep Curl | Adjustable Dumbbells |
| Tricep Pushdown | Cable Machine |
| Shoulder Press | Adjustable Dumbbells |
| Lat Pulldown | Cable Machine |
| Leg Press | Squat Rack |
| Chest Fly | Adjustable Dumbbells |
| Bench Press | Barbell |
| Push Up | Barbell |
| Push Up | Barbell |

10 rows (Limited to only 100 rows)

GROUP BY clause with aggregate functions

```sql
SELECT muscle_group, COUNT(*) AS total_exercises
FROM Exercises
GROUP BY muscle_group;
```

| muscle_group | total_exercis |
| --- | --- |
| Upper Chest | 1 |
| Chest | 3 |
| Triceps | 1 |
| Shoulders | 2 |
| Back | 1 |
| Biceps | 1 |
| Legs | 1 |

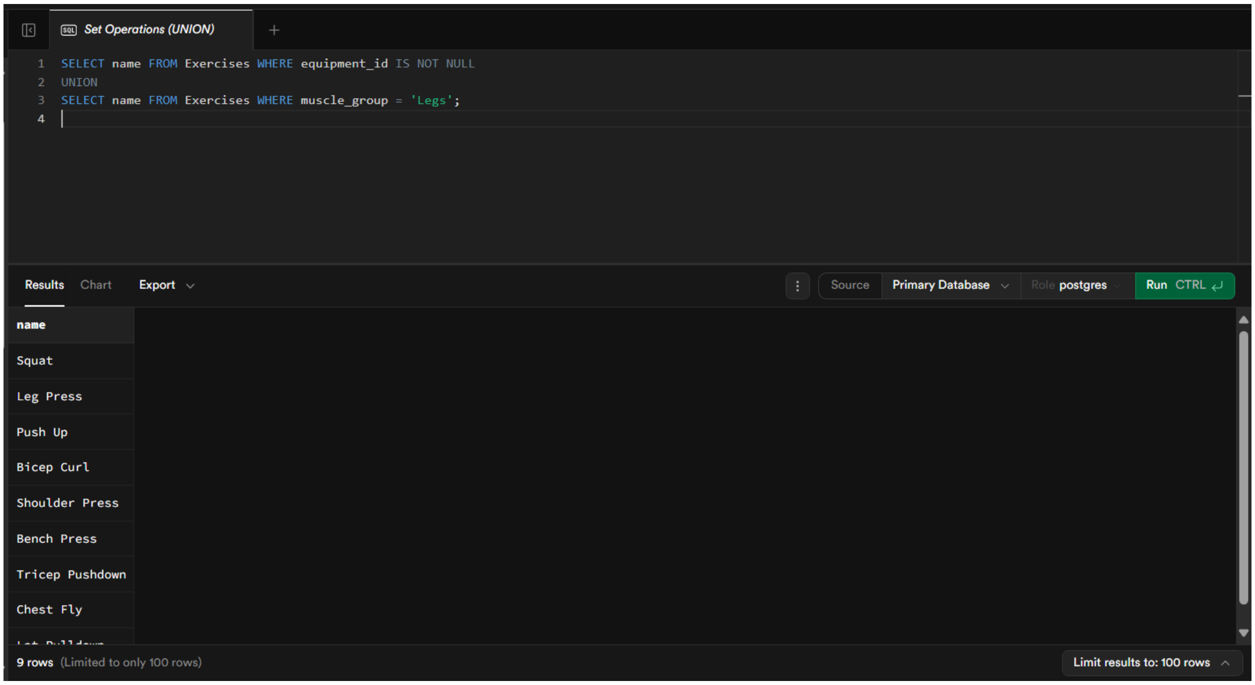
7 rows (Limited to only 100 rows)

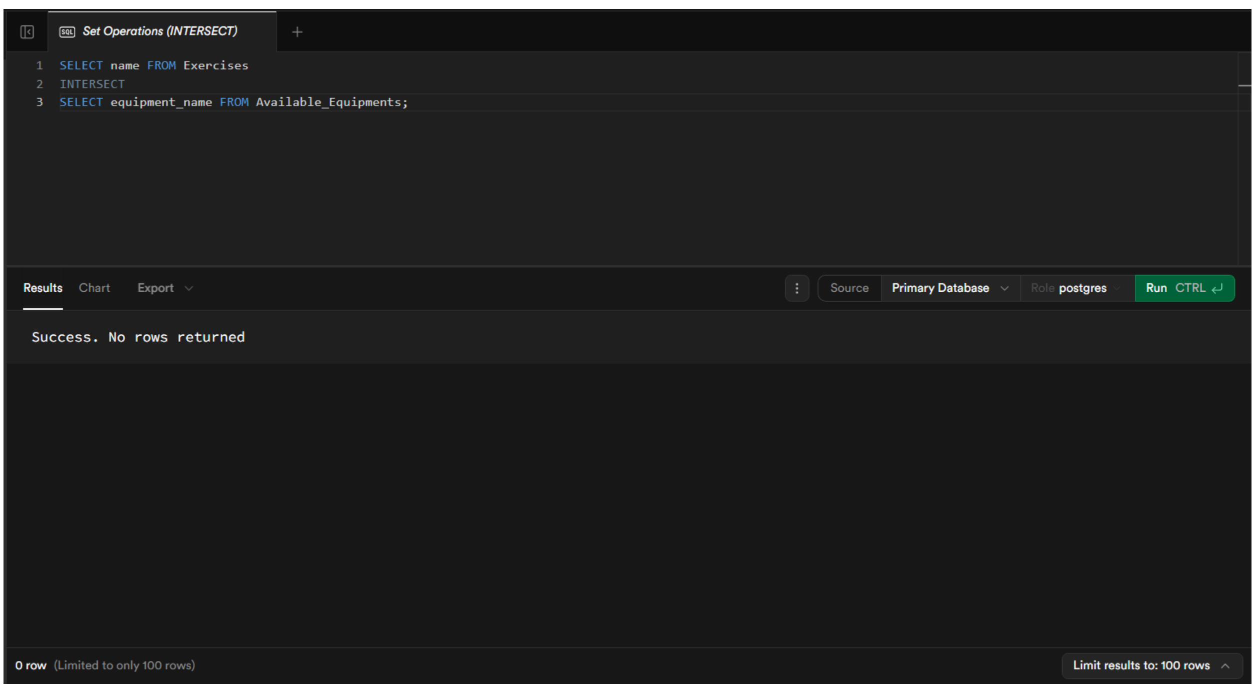
HAVING clause

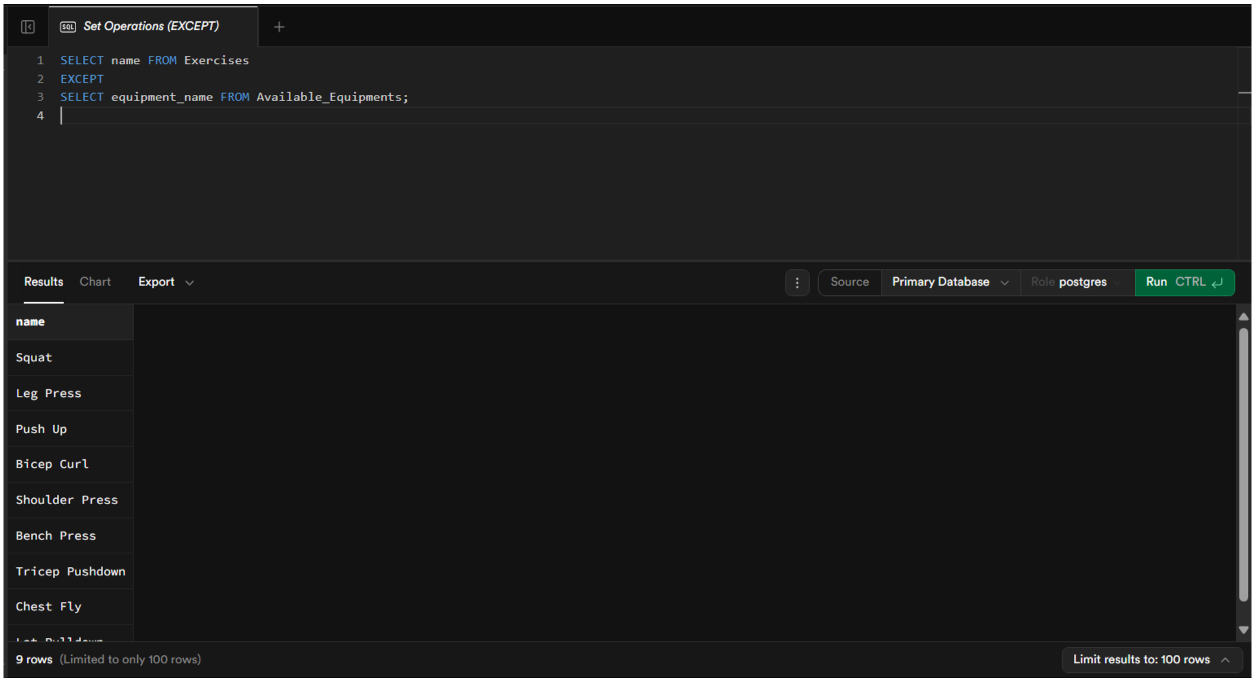
```sql
SELECT muscle_group, COUNT(*) AS total_exercises
FROM Exercises
GROUP BY muscle_group
HAVING COUNT(*) > 2;
```

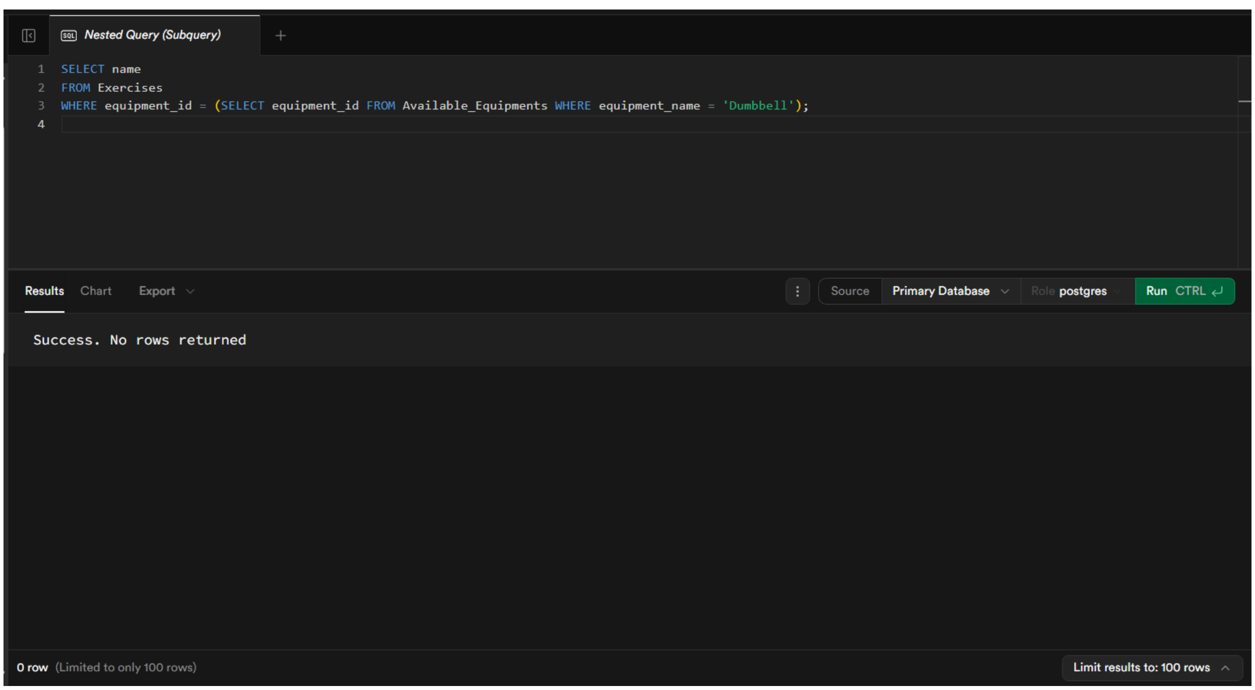
| muscle_group | total_exercis |
| --- | --- |
| Chest | 3 |

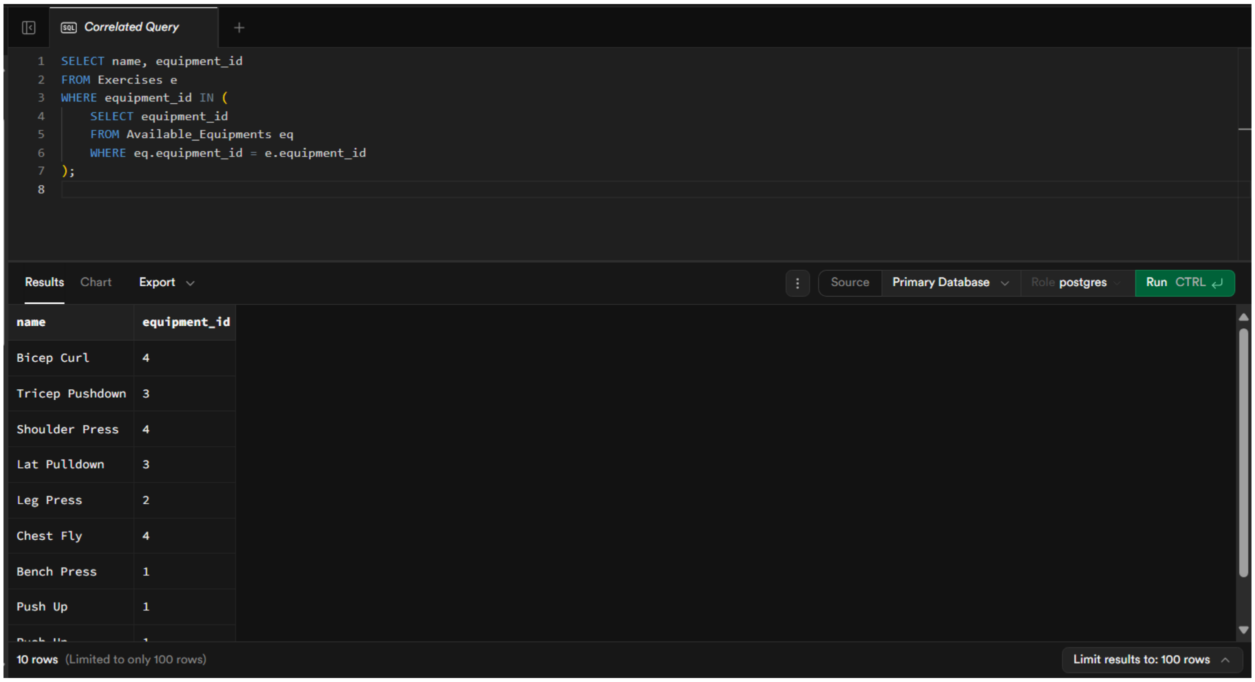
1 row (Limited to only 100 rows)


Set Operations (UNION)
SELECT name FROM Exercises WHERE equipment_id IS NOT NULL
UNION
SELECT name FROM Exercises WHERE muscle_group = 'Legs';
Results
Chart
Export
Source
Primary Database
Role postgres
Run CTRL
name
Squat
Leg Press
Push Up
Bicep Curl
Shoulder Press
Bench Press
Tricep Pushdown
Chest Fly
9 rows (Limited to only 100 rows)
Limit results to: 100 rows


Set Operations (INTERSECT)
SELECT name FROM Exercises
INTERSECT
SELECT equipment_name FROM Available_Equipments;
Results
Chart
Export
Source
Primary Database
Role postgres
Run CTRL
Success. No rows returned
0 row (Limited to only 100 rows)
Limit results to: 100 rows


Set Operations (EXCEPT)
SELECT name FROM Exercises
EXCEPT
SELECT equipment_name FROM Available_Equipments;
Results
Chart
Export
Source
Primary Database
Role postgres
Run CTRL
name
Squat
Leg Press
Push Up
Bicep Curl
Shoulder Press
Bench Press
Tricep Pushdown
Chest Fly
9 rows (Limited to only 100 rows)
Limit results to: 100 rows


Nested Query (Subquery)
SELECT name
FROM Exercises
WHERE equipment_id = (SELECT equipment_id FROM Available_Equipments WHERE equipment_name = 'Dumbbell');
Results
Chart
Export
Source
Primary Database
Role postgres
Run CTRL
Success. No rows returned
0 row (Limited to only 100 rows)
Limit results to: 100 rows



Correlated Query
SELECT name, equipment_id
FROM Exercises e
WHERE equipment_id IN (
SELECT equipment_id
FROM Available_Equipments eq
WHERE eq.equipment_id = e.equipment_id
);
Results
Chart
Export
Source
Primary Database
Role postgres
Run CTRL
name
equipment_id
Bicep Curl
4
Tricep Pushdown
3
Shoulder Press
4
Lat Pulldown
3
Leg Press
2
Chest Fly
4
Bench Press
1
Push Up
1
10 rows (Limited to only 100 rows)
Limit results to: 100 rows